P1 Nov 2011

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1.

1. $F = 6\pi\eta v r \Rightarrow \eta = \frac{F}{6\pi v r}$

6π - no units
F - ma - $kg\,m\,s^{-2}$
v - $m\,s^{-1}$
r - m

$\eta = \frac{kg\,m\,s^{-2}}{m\,s^{-1}\,m}$

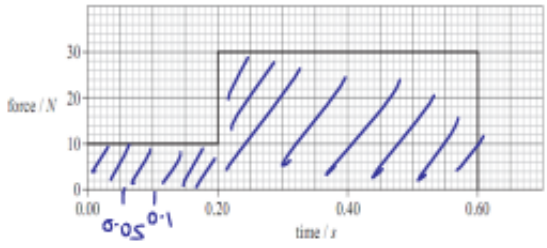
$= kg\,m^{-1}\,s^{-1}$ (C)

2.

2.  $20\,m\,s^{-1}$

$d = vt$
$= 20 \times 2.5$
$= 50\,m$ (D)

3.

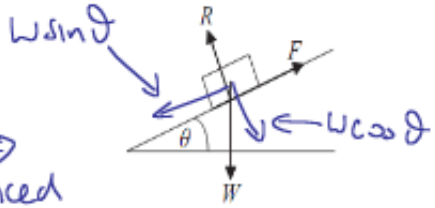
3. 

Force × time = change in momentum
= area under graph = $28 \times 0.05 \times 10$
= $14\,N\,s = \Delta mv$

$\Delta v = \frac{\Delta mv}{m} = \frac{14}{2} = 7\,m\,s^{-1}$ (A)

4.

4.



Constant speed ⇒ Forces balanced

Taking components parallel to slope

→ $W\sin\theta = F$

D

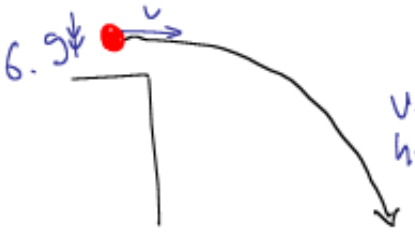
5.

5. If travelling in a circle force is towards the centre



D

6.



Vertical increases with g
Horizontal constant

B

7.

7. $\frac{GMm}{r^2} = \frac{1}{2}mv^2$ $v = \sqrt{\frac{2GM}{r}}$

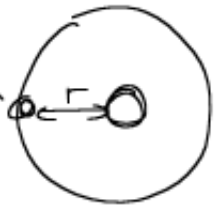
r – radius of Earth
M – mass of earth
G – constant

B

8.

8. $\frac{GMm}{r^2} = \frac{mv^2}{r}$

$\Rightarrow v = \sqrt{\frac{GM}{r}}$



reduce radius $\Rightarrow$ increase speed

$PE = \frac{GM}{r}$ $\therefore$ closer $\Rightarrow$ smaller PE

D

9.

9. 273K − 0°C

$\Rightarrow$ 350K − (350 − 273)°C

= 77 °C

C

10.

10. 12g of ^{13}C = ½ moles so

same as 6g of ^{12}C

A

11.

11. Entropy is a measure of a systems disorder.

C

12.

12. $PV = nRT \Rightarrow T = \frac{PV}{nR}$

so if 2P and V/2 $T = \frac{PV}{nR}$

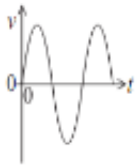
no change

B

1

13.

13.



acceleration = $\frac{dv}{dt}$

gradient of v-t

⇒ starts at the max



B

14.

14. gamma → UV → visible → IR → micro → radio

C

15.

15. ½ cycle out of phase = π C

16.

16. eeeee oooouw



A

17.

17. Diffraction is a WAVE phenomenon D

(some of these are quite easy)

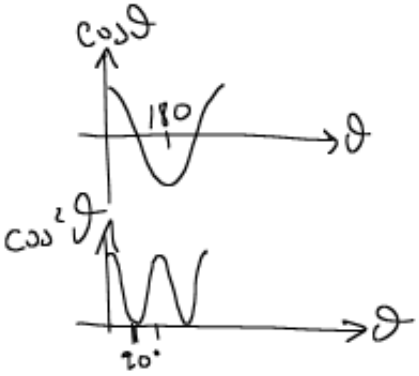
18.

18. $\omega = \frac{2\pi}{T}$ so $v = v_0 \sin \omega t$
$= v_0 \sin\left(\frac{2\pi}{T}\right)t$ (A)

19.

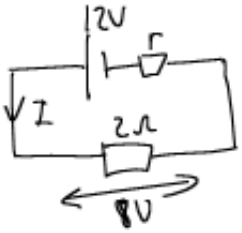
19. $I = I_0 \cos^2\theta$

(D)



20.

20. With switch open meter → EMF
⇒ EMF = 12V



from pd. across 2Ω resistor
$8 = I \times 2$
$\Rightarrow I = 4\,A$

for whole cct. $\mathcal{E} = IR + Ir$
$\Rightarrow 12 = 8 + Ir$
$\Rightarrow 4 = Ir = 4r$
$r = 1\Omega$

(A)

21.

21. $E = VIt$ but $I = V/R$
$\Rightarrow E = \frac{V^2}{R} \times t$

(D)

22.

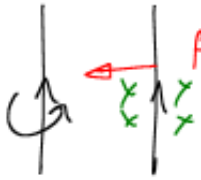
22. -ve ⇒ +ve test charge is attracted.
No field inside



B

23.

23. Wires attract each other ⇒ Y to left



A

24.

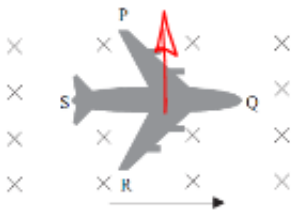
24. $V_{rms} = \frac{V_{peak}}{\sqrt{2}} = \frac{100}{\sqrt{2}}$ (something between 1 and 2)
⇒ between 50 & 100V
∴ 70V

D

25.

25. Fleming RHR
R → P

B



26.

26. most not deflected
some bounce back (a few)
some deflected

C

27.

27. $^{235}_{92}U + n \rightarrow ^{141}_{56}Ba + ^{A}_{Z}Kr + 3n$

nucleon no. $235 + 1 = 141 + A + 3$

$A = 92$

proton no. $92 = 56 + Z \Rightarrow Z = 36$ (A)

28.

29. $\lambda = \frac{h}{p} = \frac{h}{mv}$

but $U_k = \frac{1}{2}mv^2$



$\Rightarrow v = \sqrt{\frac{2Ue}{m}}$

$mv = m\sqrt{\frac{2U_k}{m}} = \sqrt{\frac{2U_k m^2}{m}} = \sqrt{2U_k m}$

$\therefore \lambda = \frac{h}{\sqrt{2U_k m}}$ (A)

29.

29. Increasing $\lambda \Rightarrow$ reduced frequency ($v = f\lambda$) so energy per photon is reduced ($E = hf$) $\Rightarrow$ KE of electrons less

Decrease $\lambda \Rightarrow$ increased $f \Rightarrow$ electrons have more KE. (B)

increase $\lambda \Rightarrow$ smaller KE. Increase I $\Rightarrow$ more photons but no change in KE.

Increasing Intensity without changing $\lambda \Rightarrow$ more photons but each photon same Energy $\Rightarrow$ more electrons but same KE

30.

30. Square of amplitude of wavefunction gives probability

D

31.

31. 

to balance charge must be β+ A

32.

32. 1 minute = 60s = 3×20s
⇒ 3 ½ lives

½ × ½ × ½ = ⅛ remaining

so ⅞ decayed. D

I bet a lot of students answered A
Yep 38%

33.

33. Absorption spectra are evidence for Electrons

B

34.

34. Energy density = Energy/mass $J kg^{-1}$

B

35.

35 from data book $P = \frac{1}{2}\rho\pi r^2 v^3$

if $r \to \frac{r}{2}$ & $v \to 2v$

$P = \frac{1}{2}\pi\left(\frac{r}{2}\right)^2(2v)^3 = \frac{1}{2}\rho\pi\frac{r^2}{4}8v^3$

$= \rho\pi r^2 v$ so $2P$

(C)

36.

36. P per $m^2 = \frac{P}{4\pi R^2} = \frac{4\times10^{26}}{4\pi(2\times10^{11})^2}$

$\sim \frac{10^{26}}{6\times10^{22}} = 1/6\times10^4 \sim 10^3 W$ (A)

37.

37. Melting glacier → less ice → greater Albedo → more radiation absorbed → global warming

Rising of sea level is a consequence but won't increase global warming (D)

Rising sea temp → emits more CO_2 → increases global warming

Deforestation → less CO_2 taken out of atmos. → increased global warming

* What a strange question

38.

38. 27 (B)

16	8	4	2	1
1	1	0	1	1

$16 + 8 + 0 + 1 + 1 = 27$

39.

39. $\frac{1}{4}\lambda = \frac{600}{4} = 150\,nm$ (A)

40.

40. $Q = CV = 100\times10^{-6}\times 2 = 200\,\mu C$ (D)

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